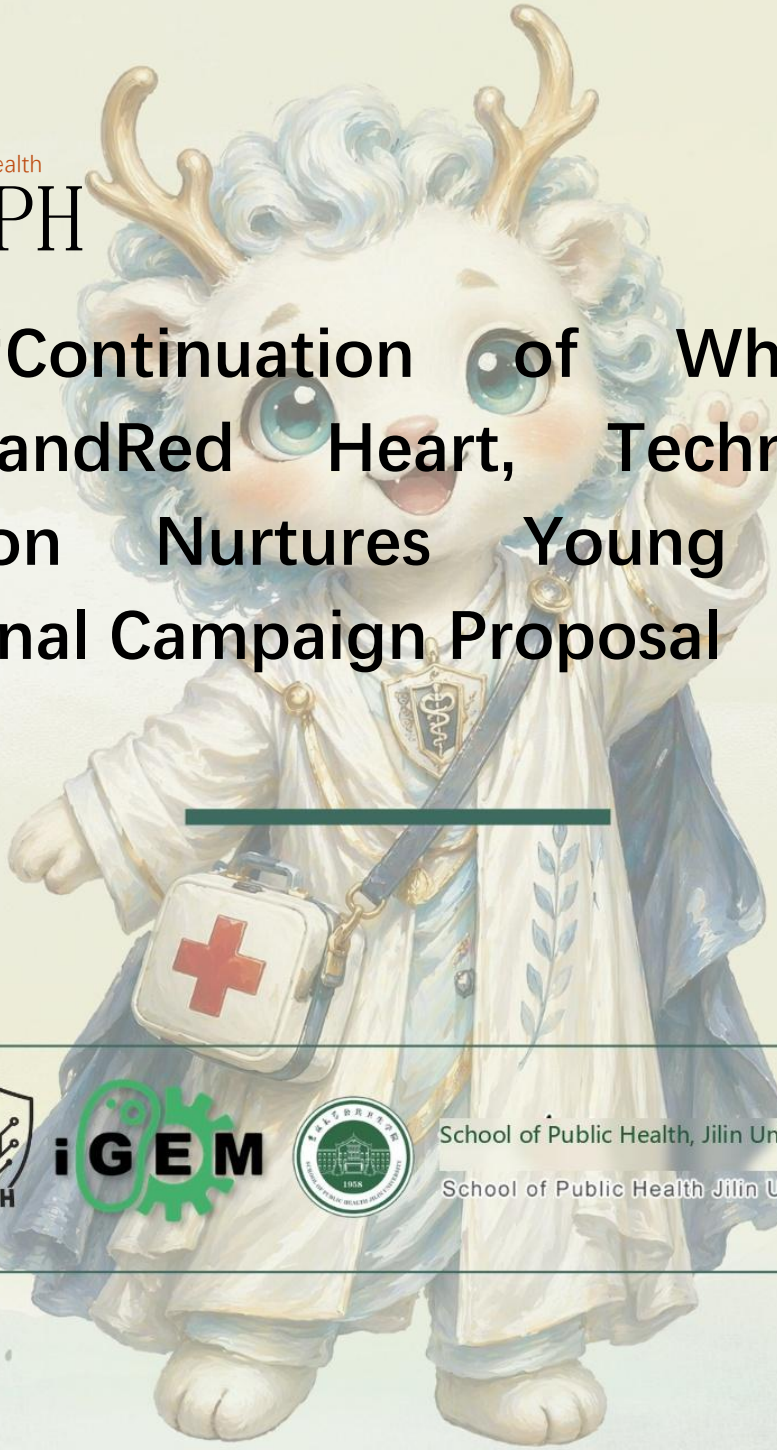
Safeguarding Life·Public Health

JLU-SPH

2026 “Continuation of White-Coat HeritageandRed Heart, Technological Innovation Nurtures Young Minds” Educational Campaign Proposal



JLU-SPH iGEM

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School of Public Health Jilin University

JILIN UNIVERSITY · SCHOOL OF PUBLIC HEALTH

1. Basic Information of the Activity

- Activity Title: "Passing on the Spirit of White Coats and Red Hearts, Inspiring Scientific Dreams in Children's Hearts" Educational Outreach
- Activity Theme: Inheriting the memory of local revolutionary medical practitioners and sowing the seeds of life science dreams among rural children
- Activity Type: ☑ Science Education and Popularization ☐ Public Welfare Services ☐ Cultural Promotion ☐ Community Services ☐ Other:
- Organizer: JLU-SPH Team

2. Objectives of the Activity

1. Core Objective: Based on the historical memory of local revolutionary medical practitioners at Gegong Primary School in Tang County, Hebei, integrate the revolutionary medical spirit, fundamental health protection knowledge, and introductory concepts of synthetic biology (合成生物学, "synthetic biology": the redesign and engineering of biological systems for new functions). Enable rural children to understand that science is not only found in textbooks, but also originates from real stories of safeguarding lives, thus advancing localized and indigenous life science education.

2. Specific Objectives:

○ Knowledge/Cognitive Level: Enable rural elementary students to gain basic understanding of pathogens (pathogen: a microorganism that causes disease), fundamental hygiene and protection knowledge, and the foundational concepts of synthetic biology. Comprehend historical stories of local revolutionary medical practitioners who protected community health, and establish connections between revolutionary spirit and modern life sciences.

○ Behavioral/Skills Level: Guide students to actively observe health phenomena in daily life, encourage them to express their imagination regarding "scientific safeguarding of life," and develop their abilities in expression, thinking, and creative imagination.

○ Emotional/Attitudinal Level: Instill a strong sense of national pride and empathy, appreciate the spirit of medical professionals who protect lives, stimulate curiosity about life sciences, and foster the awareness that "science is all around us."

3. Expected Outcomes:

○ Target Coverage: Approximately 40-50 participants.
○ Satisfaction Goal: Above 90%
○ Completion of a comprehensive activity process record

3. Target Audience

- Core Audience: Rural elementary students at Gegong Primary School, Qijiazuo Township, Tang County, Hebei Province
- Audience Profile: Located in a historical revolutionary area, with local red history and cultural background; limited access to cutting-edge life science popularization resources; high imagination, skilled at understanding knowledge through stories and concrete examples; ideal candidates for science enlightenment using indigenous historical stories as entry points.

4. Activity Time and Location

- Activity Date: July 17, 2026
- Total Activity Duration: Approximately 40 minutes

- Activity Venue: Classroom at Gegong Primary School, Qijiazuo Township, Tang County, Hebei Province

5. Activity Content and Procedures

[Overview of Activity Process]

Stage	Time	Session Title	Detailed Content	Responsible Personnel	Required Materials/Key Filming Points
Stage	20min	Preparatory Stage: Venue and Materials	Arrival at school; testing projection and audio equipment; distributing student task sheets; setting up the venue; communicating with school teachers	Main Speaker, Teaching Assistants	Popular Science PPT
Main Stage	40min	Life Science Popularization	Explain knowledge of pathogens, basic protection measures, and synthetic biology exploration in terms easily understood by children	Main Speaker, Teaching Assistants	PPT, scoreboard, cultural and creative prizes
Concluding Stage	30min	Creation, Summary, and Resource Delivery	Students complete creative sharing; activity summary; deliver a complete open-source teaching resource package to the school; collect oral and written feedback from students	All Members	PPT

[Detailed Explanation of Each Session]

●Session 1: Life Science Popularization + Interactive Sharing

- **Core Content:** Begin from the authentic local historical memory of revolutionary medical practitioners in Gegong, narrating stories of doctors overcoming challenging conditions to safeguard the health of rural residents during the revolutionary era. Establish a dialogue between "medical practitioners of the past" and "modern life sciences," serving as an emotional entry point for science education. Avoid complex technical terms, use familiar metaphors to explain what pathogens are and how to implement daily protective measures; brief introduction to synthetic biology—humans modifying components of life much like "building blocks" to protect health.

- **Key Talking Point:** "How do you hope science will safeguard our lives in the future?"

●Session 2: Student Task Sheet Creation, Summary, and Open Resource Delivery

- **Core content:** Students write or draw their own concepts of "science safeguarding life" on the task sheet; the team delivers a complete set of open-source teaching resource packages onsite to Ge Gong Primary School, which include localized PPT materials, a teacher's guide, student task sheets, and activity proposals. This facilitates the autonomous reuse and implementation of the course by school teachers in the future.
- **Key Points:** Conduct a brief activity summary and collect feedback on the activity

6. Personnel Assignment and Roles

●General Supervisor (1 person): Responsible for overall external communication with Gegong Primary School, coordinating itinerary and on-site procedures, managing unexpected issues.

●Main Speaker (1 person): Skilled in explaining concepts to elementary students in a simple manner, responsible for classroom introduction, science teaching, and interaction.

●On-site Teaching Assistants (3 persons): Distribute materials and maintain order.

●Photography and logistics personnel (2 people): Material transportation, onsite image documentation, compilation and printing of open-access educational resource package, and post-event venue cleanup.

VII. Inventory List

- Equipment Category:
  - Notebook computer, projector
  - Camera/high-definition mobile phone
- Educational Category:
  - Popular Science PPT
  - Student task sheets
  - Activity feedback questionnaire
- Logistics Category:
  - Standard first aid kit
  - Cleaning wet wipes, garbage bags

VIII. Risk Contingency Plan

- Personnel risk: crowding and minor injuries (abrasions and bruises) among students during exhibition viewing. Immediately evacuate the crowd, use the first aid kit for onsite treatment, and promptly notify the class teacher.
- Venue risk: venue temporarily unavailable. Activate backup classroom, and confirm alternative venues with the school administration in advance.
- Equipment risk: buzzer or projector malfunction; switch to manual hand-raising response, and use backup computer plus handwriting tablet to assist explanation.

- **IX. Effectiveness Evaluation**
- Process evaluation: onsite observation, procedural documentation, assessment of correct response rate.
- Satisfaction evaluation: collect paper feedback questionnaires, conduct random onsite verbal interviews with students, and gather responses immediately after the event.